Coarse-Graining

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1 Notations and Preliminaries

We follow the standard notations commonly used in the literature. For an integer n, we denote by $[n]$ the set of integers $\{1, 2, \ldots, n\}$. To present a countable set, which might or might not be finite, where the number of elements does not matter, we use curly brackets with a subscript denoting the first index of the set. For example, $\{u_i\}_{i=1}$ represents the set $u_1, u_2, \ldots$. For finite sets, we might add a superscript to present the size of the set. If the first index is clear from the context, then we might omit the assignment. We use a colon inside curly brackets to specify a set of elements satisfying a condition, i.e., for sets A, B, the notation $\{a : a \in A, a \in B\}$ equals $A \cap B$. Curly brackets without subscripts or conditions refer to singletons. For example, for a given integer i, the set $\{u_i\}$ is the set that contains only the element u_i. The size of a set A is denoted by $|A|$. For an integer i, we denote by $\mathbf{1}_i$ the vector consisting of one at the ith coordinate and zero elsewhere. The char $|$ symbolizes a restriction, for example $x|_y$, is the restriction of the 'entity x' to 'y'. The particular behavior of the restriction would change depending on the target type.

When not otherwise mentioned, we consider quantum states over qubits, each lying in a two-dimensional Hilbert space $\mathcal{H}_2$, and denote by $\mathcal{H}_{2^n}$ the Hilbert space of an n-qubit system. The $\otimes$ operator stands for tensor product, for example: $\mathcal{H}_{2^n} = \mathcal{H}_2^{\otimes n}$. The space of linear operators on the Hilbert space is denoted by $L(\mathcal{H})$. Pure quantum states and their dual presentations are denoted by $|\psi\rangle, \langle\psi|$, and in the matrix representation, we use $|\psi\rangle\langle\psi|$ to denote its density matrix. We will use ρ to denote a mixed state. The action of an operator A on a mixed state is written as $A \circ \rho$. For example, if A is a unitary operator, then $A \circ \rho = A\rho A^\dagger$. The letters, X and Z, will usually denote the single qubit X and Z Pauli operators. Yet, where the Pauli matrices are out of context, we use these letters for other variables. The Pauli group itself, over n qubits, is denoted by $\mathcal{P}^{\otimes n}$, and when the number of qubits is either clear or doesn't matter, we omit the superscript. For a Pauli $P \in \mathcal{P}$, we denote by $X(P)$ and $Z(P)$ the X- and the Z-components of P such that $P = X(P)Z(P)$. The gates 'Controlled Not' and 'Controlled Z' are denoted by **CX** and **CZ**, and the gates Hadamard, S, and T are denoted by:

$$\mathbf{H} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}, \mathbf{S} = \begin{bmatrix}1 & 0\\ 0 & i\end{bmatrix}, \mathbf{T} = \begin{bmatrix}1 & 0\\ 0 & e^{i\frac{\pi}{4}}\end{bmatrix}$$

For a quantum gate U, we denote by $|U\rangle$ the resource state that can be used to simulate U. Notice that, in general, the resource state is not unique. In this work, we address specific states that will be discussed later. The function $\mathbf{Pr}[\,\cdot\,]$ is the probability function. In the language of groups, for a group $\mathcal{G}$ and elements $g_1, \ldots, g_l$, we denote by $\langle\{g_1, \ldots, g_l\}\rangle$ the subgroup generated by $g_1, \ldots, g_l$.

Throughout the paper, we will deal with CSS codes [CS96], [Ste96]. For a linear subspace A, we use $|A\rangle$ to denote the uniform superposition over the vectors in A. A CSS code will usually be denoted by the pairs $\mathcal{C}_X, \mathcal{C}_Z$ such that the Z-stabilizers match the codewords of $\mathcal{C}_Z^\perp$, and the logical codewords in the computational basis are the elements in $\mathcal{C}_X/\mathcal{C}_Z^\perp$. The logical zero codeword is $\left|\mathcal{C}_Z^\perp\right\rangle$. We denote the equivalence

relation up to Z-stabilizer addition by $\sim_{\mathcal{C}_X}$. Namely, for $x_1, x_2 \in \mathbb{F}_2^n$, we have $x_1 \sim_{\mathcal{C}_X} x_2$ if $x_1 + x_2 \in \mathcal{C}_Z^{\perp}$. By the same way we define $\sim_{\mathcal{C}_Z}$.

The Computation Model. Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, at different space-time locations. Each such location records when the gate is applied and on which qubits it acts. Formally:

Definition 1.1 (Quantum Circuit)**.** *A* space-time location *is a pair $l = (t, S)$, where t is a time coordinate and $S = (s_1, \ldots, s_\Delta)$ is an ordered tuple of qubit indices. For a finite collection of quantum gates*

$$\mathcal{G} \subset \left\{ g \colon L(\mathcal{H}_2^{\otimes \Delta}) \to L(\mathcal{H}_2^{\otimes \Delta}) \right\},$$

and integers $w, d \in \mathbb{N}$, a quantum circuit C of depth d and width w is a collection of tuples

$$C = \{u_i = (g_i, l_i)\}_i,$$

where each $g_i \in \mathcal{G}$ and each location $l_i = (l_{i,1}, l_{i,2})$ belongs to $[d] \times [w]^\Delta$. We call the tuples u_i the gates of C. They must satisfy the following compatibility condition: if two gates have the same time coordinate, then their supports are disjoint. Namely, for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies that the sets of qubit indices appearing in $l_{i,2}$ and $l_{j,2}$ are disjoint. We denote by

$$\mathrm{Loc}(C) := \{l_i : u_i = (g_i, l_i) \in C\}$$

the set of locations occupied by the gates of C. We define the t-layer of C to be its restriction to gates at time t:

$$C|_t := \{u_i = (g_i, l_i) \colon u_i \in C \text{ and } l_{i,1} = t\}.$$

Later, we will restrict $\mathcal{G}$ to contain $\mathbf{X}, \mathbf{Z}, \mathbf{CX}, \mathbf{H}, \mathbf{S}, \mathbf{T}$, and measurements. To define the action of a quantum circuit C on a given mixed state, we first associate C with a directed acyclic graph.

Definition 1.2 (Associated Computation DAG)**.** *Let $C = \{u_i\}_i$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the directed acyclic graph $\pi_C = (\mathcal{U}, E)$ as follows:*

1. *The vertex set is the gate set of C, namely $\mathcal{U} = \{u_i\}_i$.*
2. *For $u_i = (g_i, l_i)$ and $u_j = (g_j, l_j)$, there is a directed edge $(u_i, u_j) \in E$ if and only if j is one of the minimizers of*

$$\left\{ l_{k,1} : (g_k, l_k) \in \mathcal{U},\ l_{k,1} > l_{i,1},\ l_{k,2} \cap l_{i,2} \neq \emptyset \right\}.$$

Equivalently, u_j is one of the earliest later gates whose support intersects the support of u_i. There may be more than one such minimizer, so the out-degree may exceed one.

Suppose that $J = (j_1, j_2, \ldots, j_k)$ is an ordering of the gate indices that agrees with a topological order induced by π_C. Then we define the associated partial circuits $C|_J = \{C|_{J,\to i}\}_{i=1}^k$ by

1. *$C|_{J,\to 1} = \{u_{j_1}\}$.*
2. *For $1 \le i < k$, set $C|_{J,\to i+1} = C|_{J,\to i} \cup \{u_{j_{i+1}}\}$.*

We call $C|_{J,\to i}$ the partial computation circuit up to step i. When the order itself does not play a role, except for being time-respecting (namely, a gate u that is applied before gate u' has a lower step than u''s step in J), we will omit J from the subscript and briefly write $C|_{\to i}$ to describe a partial circuit up to that step.

Definition 1.3. *Let ρ be a density matrix over w qubits, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}_{i=1}$ on ρ, according to a topological order induced by π_C.*

We say that $\rho_1, \rho_2, \ldots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u'_i \circ \rho_i$, where $\{u'_i\}_{i=1}$ is a prefix of a topological ordering of $\{u_i\}_{i=1}$. If $\{u'_i\}_{i=1}$ contains all the gates in $\{u_i\}_{i=1}$, then we call $\rho_1, \rho_2, \ldots, \rho_k$ a running.

For a quantum circuit C, we would like to discuss its action in the presence of different outcomes of faults. The following definition encodes the running in the presence of a specific set of faults $\mathcal{E}$.

Definition 1.4 (C subjected to $\mathcal{E}$ every m computational layers)**.** *Let $C = \{u_i\}_i$ be a quantum circuit with depth d and width w, let $m \in \mathbb{N}$, and set*

$$d_m := \left\lceil \frac{d}{m} \right\rceil .$$

Let $\mathcal{L} \subset [d_m] \times [w]$ be a set of locations, and let $\mathcal{E}$ be a quantum circuit, such $Loc(\mathcal{E}) = \mathcal{L}$. We name $\mathcal{E}$ subsets of faults, and we name the layers of $\mathcal{E}$ fault layers. The circuit C subjected to $\mathcal{E}$ every m computational layers, denoted by $C[\mathcal{E}]_m$, is obtained by setting the first layer to be the first layer of $\mathcal{E}$, then placing in order the layers of C and inserting one fault layer after every m consecutive C's layers. Thus the second fault layer is inserted after the layers $1, \ldots, m$ of C, the third is inserted after the layers $m+1, \ldots, 2m$ of C, and so on. Concretely, each gate $u_i = (g_i, (t_i, S_i))$ of C is moved to the time coordinate

$$t_i + \left\lfloor \frac{t_i - 1}{m} \right\rfloor ,$$

while each fault gate $(g, (t, S)) \in \mathcal{E}$ is placed at time coordinate

$$t(m+1).$$

When $m = 1$, the gates occupy the even time coordinates and the fault layers occupy the odd time coordinates. When the parameter is omitted, we mean $m = 1$. We denote the associated computation DAG of $C[\mathcal{E}]_m$ by $\pi_{C[\mathcal{E}]}$.

Remark 1.5. *Formally, a fault circuit is just a particular quantum circuit whose gates are interpreted as faults rather than as computational operations. Thus the distinction between a quantum circuit and a fault circuit is semantic rather than structural: both are circuits in the sense of Definition 1.1, but they play different roles in the analysis.*

Example 1.6. *When $m = 3$, the 'computational' layers are sent to the times*

$$2, 3, 4, 6, 7, 8, 10, 11, 12, \ldots,$$

while the fault layers are sent to the times

$$1, 5, 9, 13, \ldots.$$

Later, all the quantum registers will be encoded via stabilizer codes. Then, following the common approach in the literature, we define the fault layers to end with stabilizer measurements, thereby forcing the faults to be Paulis (see [Got97]). Hence, we confine ourselves to noise channels of only this type.

Definition 1.7 (Noise channel and local stochastic noise channel)**.** *A noise channel $\mathcal{N}_\mathcal{E}$ is a distribution over sets of faults and their locations. Let $p \in (0, 1)$, we say that the noise channel $\mathcal{N}_\mathcal{E}$ is local stochastic with rate p if for every finite set of locations $S \subset \mathbb{N} \times \mathbb{N}$,*

$$\mathbf{Pr}_{\mathcal{E}\sim\mathcal{N}_\mathcal{E}}\left[S \subset \mathrm{Loc}(\mathcal{E})\right] \leq p^{|S|}.$$

Through this work, the sampled faults will be assumed to be Pauli.

So, $N_\mathcal{E}$ defines a probability space. The set of faults $\mathcal{E}$ is an atomic outcome, and for a given circuit C, the subjected circuit $C[\mathcal{E}]$ is a random variable. Equipped with this notation, we are allowed to assume having constant depth cycles that don't observe faults.

Definition 1.8. *Let C be a quantum circuit that contains l measurements. For possible outcomes $\alpha \in \mathbb{F}_2^l$, let $C[, \alpha]$ be the circuit obtained by conditioning that $\alpha_1, .., \alpha_l$ were measured. We extend the notation for the subjected circuit by denoting $C[\mathcal{E}, \alpha]$ as the subjected circuit where the measurements are assumed to outcome $\alpha_1, .., \alpha_l$.*

There might be multiple different ways to present a conditioned circuit with a collection of gates and their locations. Since we would like to keep the step-order invariant to a conditioning of the measurement outcome, we choose a representation that substitutes the original gates with gates that act on the same location. The following remark explains how to do that explicitly.

Remark 1.9. *Consider the presentation of a conditioned circuit as a subset of gates acting on locations. For an outcome $\alpha \in \mathbb{F}_2^{(\cdot)}$ for which a conditional gate (for example, a* **CCZ***, where the control bits are conditioned on measurement outcomes) in the original circuit C is mapped to a Pauli in $C[,\alpha]$, we substitute this gate with the Pauli on the target tensored with identity acting on the control locations. Thus, the order of the steps, the locations of each gate, and the computation graph of $C[,\alpha]$ are the same as for C (and independent of the outcomes). We summarize this by briefly saying that the locations are invariant to the conditioning of measurements.*

In our case, the conditioned circuit $C[\mathcal{E},\alpha]$ is a Clifford circuit; in particular, it sends Paulis to Paulis. That brings us to define, for a given set of faults $\mathcal{E}$ and outcomes α, a Pauli-path that describes the evolution of the faults through the circuit.

Definition 1.10 (Propagation)**.** *Let $C[\mathcal{E}]$ be a subjected quantum circuit at depth μ, and let $\alpha \in \mathbb{F}_2^{(\cdot)}$ be a possible outcome vector for $C[\mathcal{E}]$. Suppose that C is a Clifford+measurements circuit, so $C[\mathcal{E},\alpha]$ is a Clifford circuit. We define the propagation of Pauli $P \in \mathcal{P}$ through the conditioned circuit $C[\mathcal{E},\alpha]$ to be the Paulis $P_1,\ldots,P_{\mu+1}$ such that $P_1 = P$ and for any $i \in [2,\mu+1]$:*

$$P_i C[,\alpha]|_{\to i} = C[\mathcal{E},\alpha]|_i P_{i-1} C[,\alpha]|_{\to i-1} = C[\mathcal{E},\alpha]|_{\to i} P_1$$

We extend the notation to step-propagation by considering the propagation of the circuit obtained from $C[\mathcal{E},\alpha]$ by flattening the circuit such that each gate is applied at the time that equals its step.

1.1 Time Coarse-Graining

The following claim states that for local stochastic noise, the subjection of every 1-layer is equivalent to the subjection of every m-layer, up to a rescaling of the noise rate. Here, by equivalence, we mean that the actions of the subjected circuits are the same[1]. We give the proof in the appendix (see, Claim 1.14).

Claim 1.11. *Let $\mathcal{N}_\mathcal{E}$ be a local stochastic noise channel with rate p, and let C be a quantum circuit whose gates have arity at most Δ. Fix an even integer c. Then there exists a local stochastic noise channel $\mathcal{N}_{\mathcal{E}_q}$ with rate*

$$q := \left(2^{H(\Delta^{-2c})\Delta^{2c}} p\right)^{\Delta^{-c}},$$

such that for any given state ρ it holds that:

$$C[\mathcal{E}]_1 \circ \rho = C[\mathcal{E}_q]_{c/2} \circ \rho$$

In particular, up to a constant depending only on Δ and c, the rate q behaves like $p^{\Delta^{-c}}$.

[1]For a given input (mixed) state ρ, they yield the same output

Say somthing about that we assume that the gates are uniteris, otherwise the propagation does not work.

In this section, we prove that when subjecting a Clifford + measurements circuit to local stochastic noise, changing the frequency at which we introduce the fault layers translates to rescaling the noise rate. For this purpose, we are abusing the notation "propagation" by reusing it to describe a different context.

Definition 1.12 (Layer propagation)**.** *Let $C = \{u_i\}_{i=1}$ be a quantum Clifford circuit. For each time coordinate t, write $U_t := C|_t$. The family of non-empty t-cuts $\{U_t\}_t$ is called the layer decomposition of C.*

Let $t < t'$ be such that U_t and $U_{t'}$ are non-empty and $U_s = \emptyset$ for every $t < s < t'$. The forward propagation of the layer U_t across the next layer $U_{t'}$ is the operation that replaces the pair $(U_t, U_{t'})$ by a pair $(U_{t'}, V_{t\to t'})$ satisfying

$$\prod_{u\in U_t} u \prod_{v\in U_{t'}} v = \prod_{v\in U_{t'}} v \prod_{w\in V_{t\to t'}} w.$$

The layer $V_{t\to t'}$ is called the propagation expression of U_t across $U_{t'}$.

More generally, if $t < t'$, the forward propagation of U_t to time t' is obtained by iterating the above operation through the successive non-empty cuts between times t and t'. The resulting propagated layer is denoted by $V_{t\to t'}$ and is again called the propagation expression of U_t to time t'. The resulting circuit has layer decomposition

$$\ldots, U_{t-1}, U_t, U_{t+1}, \ldots, U_{t'}, U_{t'+1}, \ldots \mapsto \ldots, U_{t-1}, U_{t+1}, \ldots, U_{t'}, V_{t\to t'}, U_{t'+1}, \ldots.$$

After propagation we update the time coordinates so that they agree with the new layer in which each gate is applied.

Definition 1.13 ($C \sim_P C'$)**.** *For quantum circuits C and C', we write $C \sim_P C'$ if one can be obtained from the other by a finite sequence of layer propagations.*

By construction, $\sim_P$ preserves the implemented channel. Indeed, if $C \sim_P C'$ and ρ is any mixed state on the input register, then

$$C \circ \rho = C' \circ \rho.$$

We state Claim 1.11 in the language of layer propagation.

Claim 1.14. *Let $\mathcal{N}_\mathcal{E}$ be a local stochastic noise channel with rate p, and let C be a quantum circuit whose gates have arity at most Δ. Fix an even integer c. Then there exists a local stochastic noise channel $\mathcal{N}_{\mathcal{E}_q}$ with rate*

$$q := \left(2^{H(\Delta^{-2c})\Delta^{2c}} p\right)^{\Delta^{-c}},$$

such that $C[\mathcal{E}]_1 \sim_P C[\mathcal{E}_q]_{c/2}$. In particular, up to a constant depending only on Δ and c, the rate q behaves like $p^{\Delta^{-c}}$.

Proof of Claim 1.11. Partition the subjected circuit $C[\mathcal{E}]_1$ into consecutive blocks of c layers. Inside each block, propagate all the fault layers to the end of the block and then groupping the resultd expressions together. Doing this block by block yields a new circuit C' in which each block of $c/2$ computational layers is followed by a single fault layer. Therefore $C' = C[\mathcal{E}']_{c/2}$ for a suitable random fault set $\mathcal{E}'$, and by construction: $C[\mathcal{E}]_1 \sim_P C[\mathcal{E}']_{c/2}$.

It remains to be proven that $\mathcal{E}'$ is distributed according to a local stochastic channel. Fix one propagation expression $V^{(i)}$ and a subset $S \subset \mathrm{Loc}(V^{(i)})$. Denote by $U_1^i, \ldots, U_c^i$ the layers for which $V^{(i)}$ is their (grouped) expression. Denote the light cone of S in $U_1^i, \ldots, U_c^i$ by $\Lambda(S)$. Since the fan is bounded by Δ, we have that $|\Lambda(S)| \leq \Delta^c |S|$, and any $S' \subset \Lambda(S)$ whose propagation covers S has size at least $|S'| \geq \Delta^{-c}|S|$, and in particular contains a subset of size exactly $\Delta^{-c}|S|$.

Now, having $S \subset \mathrm{Loc}(V^{(i)})$, implies that there is a minimal S' (the minimality is regarding inclusion) such that $S' \subset \mathrm{Loc}(\mathcal{E})$. Denote by $\mathcal{W}(S)$ the set of such minimal S''s, and observe that its size is at most:

$$|\mathcal{W}(S)| \leq \binom{\Delta^c |S|}{\Delta^{-c}|S|}$$

Hence, using the fact that $\mathcal{E}$ is an outcome of local stochastic noise channel with a noise rate p we have:

$$\begin{aligned}
\mathbf{Pr}\big[S \subset \mathrm{Loc}(V^{(i)})\big] &= \sum_{\alpha} \mathbf{Pr}\big[S \subset \mathrm{Loc}(V^{(i)})|\alpha\big]\mathbf{Pr}\big[\alpha\big] \\
&\leq \sum_{\alpha} \mathbf{Pr}\Big[\bigcup_{S' \in \mathcal{W}(S|\alpha)} \{S' \subset \mathrm{Loc}(\mathcal{E})\}|\alpha\Big]\mathbf{Pr}\big[\alpha\big] \\
&\leq \sum_{\alpha} \sum_{S' \in \mathcal{W}(S|\alpha)} \mathbf{Pr}\big[S' \subset \mathrm{Loc}(\mathcal{E})|\alpha\big]\mathbf{Pr}\big[\alpha\big] \\
&\leq \binom{\Delta^c |S|}{\Delta^{-c}|S|} p^{\Delta^{-c}|S|} \leq \left(2^{H(\Delta^{-2c})\Delta^{2c}} p\right)^{|S|/\Delta^c} = q^{|S|}.
\end{aligned}$$

□

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